CMOS NAND Gate Layout Design

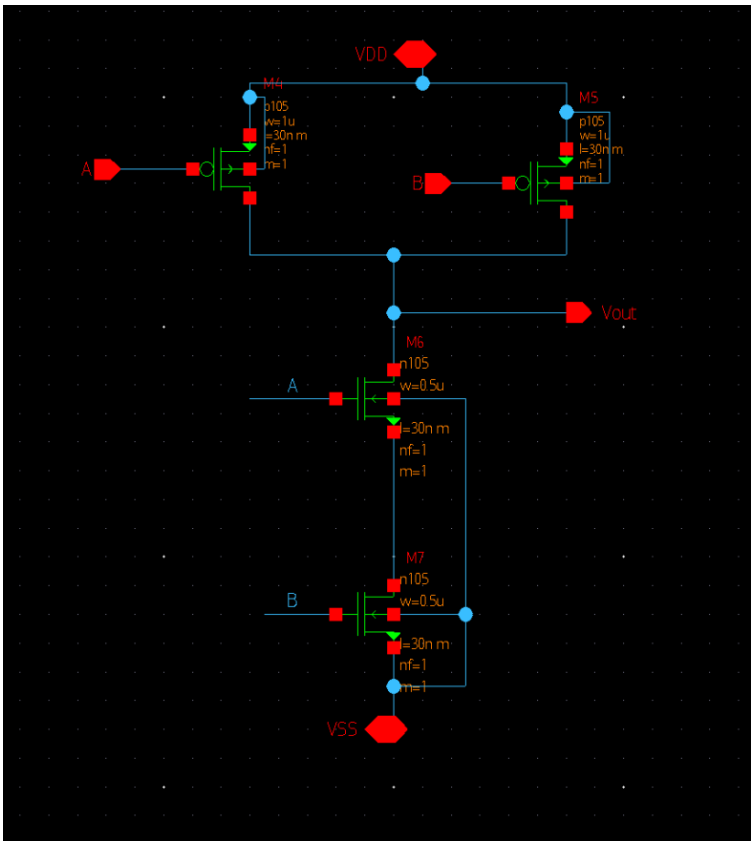
Physical Verification and Layout Parasitic Extraction

Technology Node: SAED 32/28nm Tools: Synopsys Custom Compiler, IC Validator

Overview

This document presents the layout implementation of a CMOS NAND gate using the SAED 32/28nm PDK in Synopsys Custom Compiler. The work includes understanding the schematic structure of the NAND gate, generating layout views from source, connecting the PMOS and NMOS devices to realize NAND functionality, and verifying the final layout using DRC, LVS, and layout parasitic extraction.

1 CMOS NAND Gate Schematic



CMOS NAND Gate Schematic

The CMOS NAND gate consists of two PMOS transistors in the pull-up network and two NMOS transistors in the pull-down network. In the schematic, the PMOS transistors

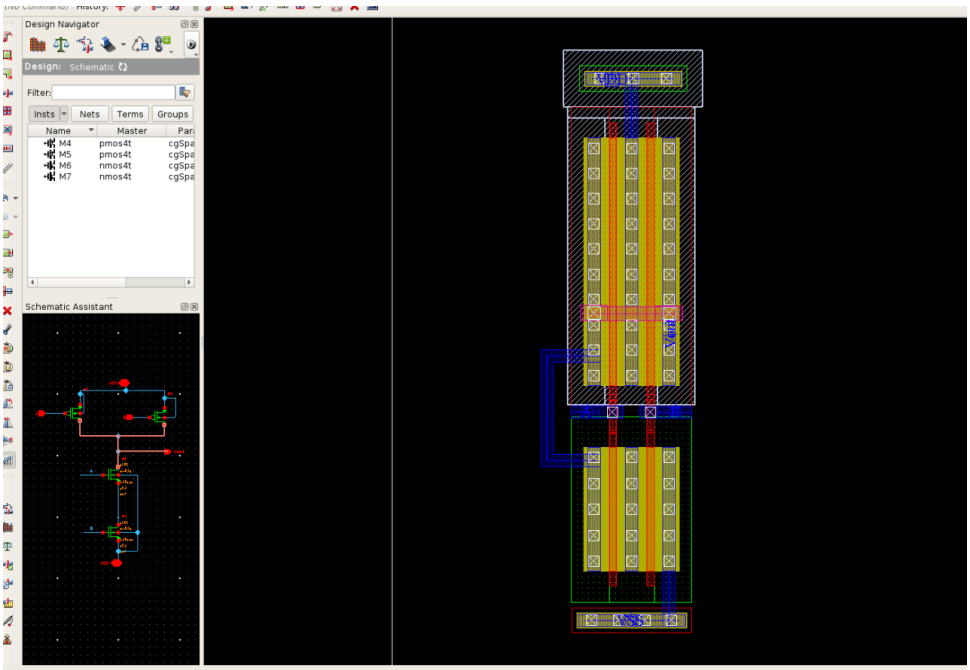
are connected in parallel, while the NMOS transistors are connected in series. This arrangement is required by the NAND truth table. The output should become low only when both inputs are high. For this reason, the NMOS transistors are placed in series so that a complete pull-down path to ground exists only when both NMOS devices are turned on. The PMOS transistors are connected in parallel so that the output can still be pulled up to VDD whenever at least one input is low.

2 Layout Generation from Source

In this NAND layout, the PMOS and NMOS devices were not created from scratch manually. Instead, the device layouts were generated from source based on the schematic and the technology information in the PDK. This approach ensures that the basic transistor geometry follows the device parameters already defined in the schematic, including transistor width and channel length.

After generating the layout from source, the main work focused on arranging the devices properly and making all required physical connections to realize the NAND function. The PMOS devices were placed in the upper region of the layout inside the N-well, while the NMOS devices were placed in the lower region. This arrangement follows the standard CMOS layout style, where the pull-up network is placed near the VDD rail and the pull-down network is placed near the VSS rail.

3 NAND Layout Integration



CMOS NAND Layout

Once the generated transistor layouts were available, the next step was to build the NAND gate by completing the physical interconnections. The two PMOS devices were positioned side by side in the upper portion of the layout. Since the PMOS network in a NAND gate is parallel, both PMOS sources were connected to the VDD rail. Their drains were then brought together and connected to a common output node. This common drain connection forms the pull-up side of the NAND output.

In the lower portion of the layout, the two NMOS devices were arranged one below the other to realize the series pull-down network. The drain of the upper NMOS was connected to the output node, while the source of the lower NMOS was connected to VSS. The intermediate nodes between the two NMOS devices were connected together to form the series path. This arrangement ensures that the output is discharged to ground only when both NMOS transistors are turned on.

The input connections were then made by routing each input to the corresponding PMOS and NMOS gates. Input A was connected to the gate of the first PMOS and the gate of the first NMOS. Input B was connected to the gate of the second PMOS and the gate of the second NMOS. This gate-level correspondence between pull-up and pull-down devices is essential to preserve the intended NAND logic.

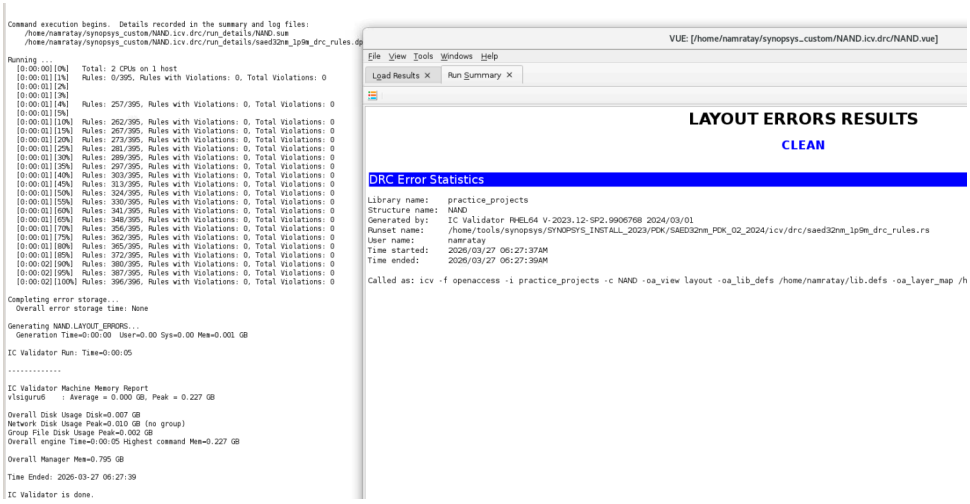
Special care was taken while routing the output node. Since the output must be common to both the pull-up and pull-down networks, the drain region of the PMOS side and the drain region of the NMOS side were routed to a single metal connection labeled as *vout*. Similarly, the upper power connection was routed to VDD and the lower power connection was routed to VSS. The resulting layout reflects the same logical connectivity present in the schematic, but in a physically valid form.

4 Connectivity Explanation

The final connectivity of the NAND layout can be understood by tracing the signal and power paths through the layout. The PMOS sources are tied to the top power rail, which represents VDD. Their drains are shorted together to form the output node. In the NMOS network, the two transistors are stacked vertically, and their shared middle connection creates the series path required for NAND behavior. The output node is taken from the drain of the upper NMOS and tied to the same common node used by the PMOS drains. The bottom of the NMOS stack is connected to VSS.

This means that when either input is low, at least one PMOS transistor remains on and provides a path from VDD to the output, making the output high. Only when both inputs are high do both NMOS transistors conduct simultaneously, completing the pull-down path and forcing the output low. The physical layout therefore preserves the exact logical operation of the schematic through correct placement and routing.

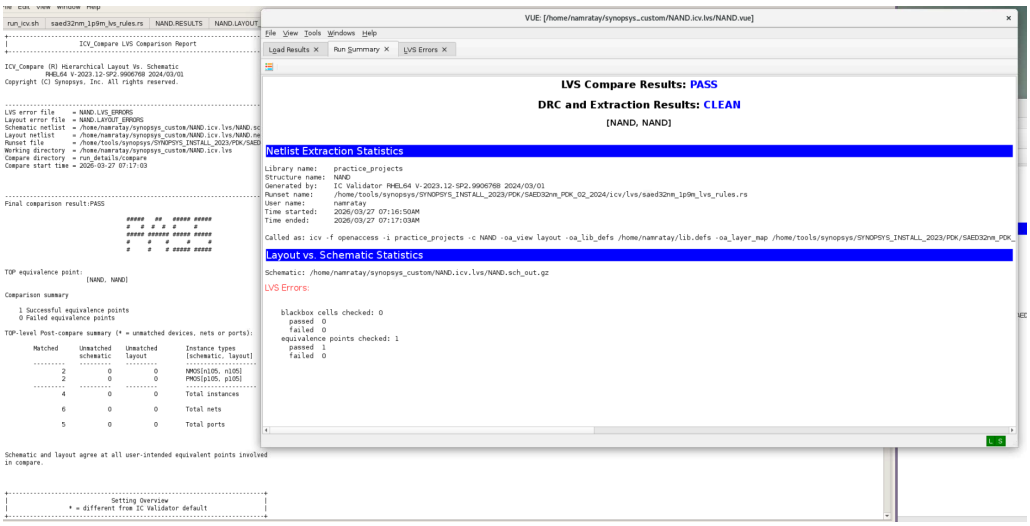
5 Design Rule Check (DRC)



DRC Clean Result

After completing the routing and finalizing the layout, design rule checking was performed using IC Validator. This step verified that the geometry of the layout satisfied the rules defined by the SAED 32/28nm PDK. The check includes spacing, enclosure, overlap, and width rules across all relevant layers. The final result was clean, indicating that the layout is physically valid and manufacturable according to the technology rules.

6 Layout Versus Schematic (LVS)

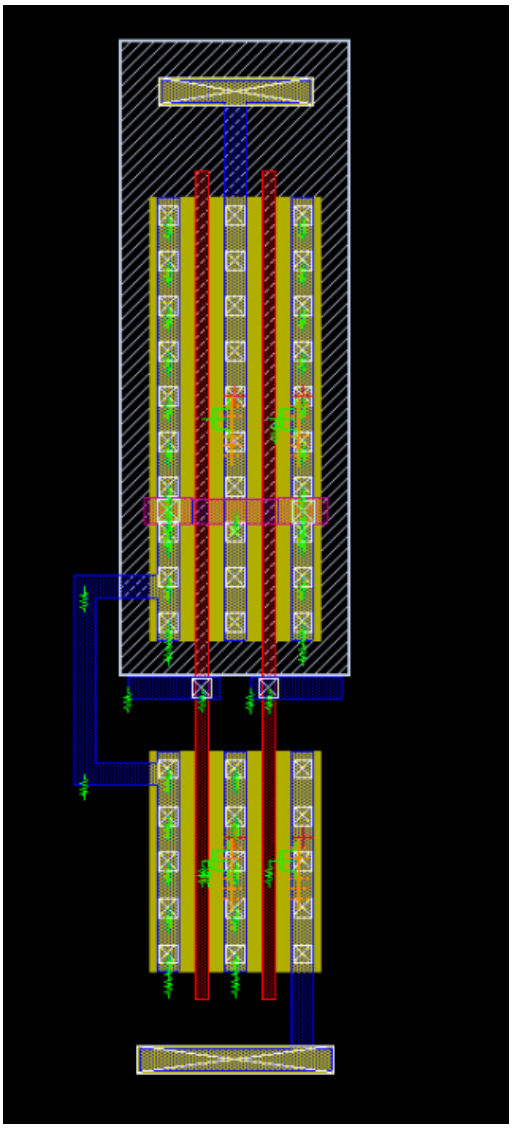


LVS Result Showing PASS

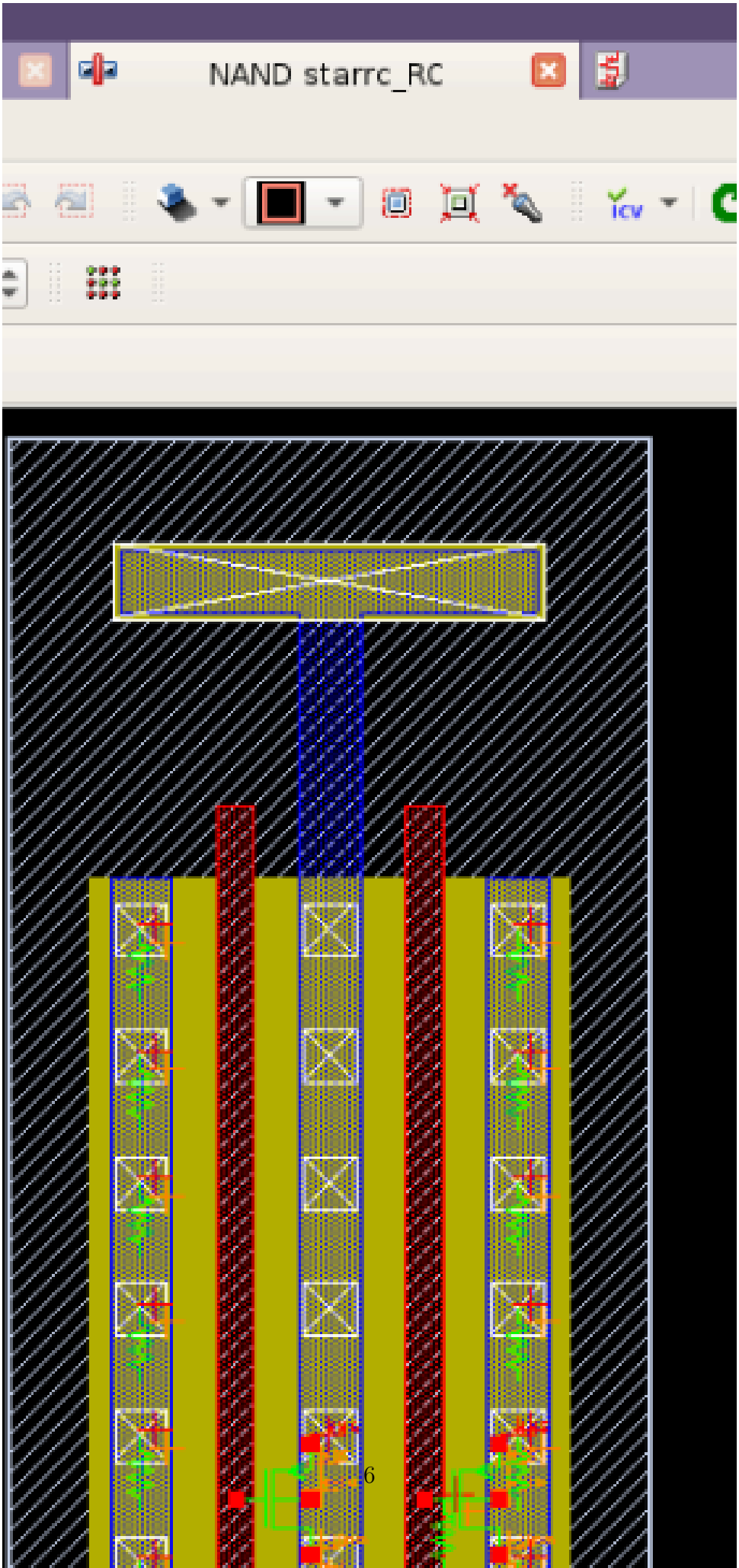
After DRC completion, LVS was performed to confirm that the physical layout matches the original NAND schematic. In this step, the netlist extracted from the layout was

compared against the schematic netlist. The result shows a successful match, which confirms that the devices, their count, and their connectivity in the layout are all consistent with the intended NAND design.

7 Layout Parasitic Extraction (LPE)



Full Layout View with Extracted Parasitic Network


NAND starrc_RC
icv

Once DRC and LVS were completed successfully, layout parasitic extraction was carried out to capture the non-ideal effects introduced by the physical implementation. In the extracted view, the layout shows parasitic elements associated with routing, contacts, and device regions. The zoomed image makes the resistance values along the interconnect paths more visible.

Parasitic resistance appears due to the finite resistance of metal lines, contacts, and diffusion regions. As routing paths become longer, the resistance increases and contributes to delay in signal propagation. Parasitic capacitance arises from the physical proximity and overlap of conductive layers, including coupling between nearby interconnects and capacitance between device terminals and substrate regions. Together, these parasitic resistance and capacitance values create RC delay, which affects the switching speed and timing behavior of the NAND gate in post-layout simulation.

8 Design Flow Summary

The NAND gate implementation followed a practical flow beginning with the schematic and pre-layout understanding of the logic structure, followed by layout generation from source and physical interconnection of the generated PMOS and NMOS devices. After completing the power, input, output, and internal series/parallel connections in the layout, the design was checked for physical correctness using DRC and for electrical equivalence using LVS. Once the layout was verified, parasitic extraction was performed to capture resistance and capacitance introduced by the physical geometry of the design.

Conclusion

The CMOS NAND gate was successfully implemented in layout by generating device layouts from source and then physically connecting them to realize the required logic function. The final layout preserves the intended parallel PMOS pull-up network and series NMOS pull-down network, passes DRC, matches the schematic in LVS, and is ready for post-layout analysis through parasitic extraction.